

$$23.1] \quad v = \frac{c}{n} = \frac{3 \times 10^8}{1.3} = \boxed{2.3 \times 10^8 \text{ m/s}}$$

$$23.3] \quad \text{Given } \lambda_{\text{air}} = 600 \text{ nm}$$

$$n_{\text{water}} = \frac{c}{v} = \frac{c}{\lambda_{\text{water}} f}, \quad f = \frac{c}{\lambda_{\text{air}}}, \quad f \text{ is the same in air and water}$$

$$n_{\text{water}} = \frac{c}{\lambda_{\text{water}}} \frac{\lambda_{\text{air}}}{c} \Rightarrow \lambda_{\text{water}} = \frac{\lambda_{\text{air}}}{n_{\text{water}}}$$

$$\lambda_{\text{water}} = \frac{600 \text{ nm}}{4/3} = \frac{3 \times 600}{4} = \boxed{450 \text{ nm}}, \quad n_{\text{water}} \text{ from table 23.1}$$

$$23.6] \quad \text{Given } f = 5 \times 10^{14} \text{ Hz} \quad n_{\text{glass}} = 1.5$$

$$(a) \quad \lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^{14}} = 6 \times 10^{-7} \text{ m} = \boxed{600 \text{ nm}}$$

$$f \text{ is same} = \boxed{5 \times 10^{14} \text{ Hz}}$$

$$\lambda_{\text{glass}} = \frac{\lambda_{\text{air}}}{n_{\text{glass}}} = \frac{600 \text{ nm}}{1.5} = \boxed{400 \text{ nm}}$$

$$23.7] \quad \text{Given } n_{\text{air}} = 1 = n_1 \Rightarrow n_2 = n_{\text{water}} = \frac{4}{3}$$

$$R = \left(\frac{n_2 - n_1}{n_2 + n_1} \right)^2 = \left(\frac{4/3 - 1}{4/3 + 1} \right)^2 = \left(\frac{1/3}{7/3} \right)^2 = \boxed{0.02} = 2\%$$

$$T = 1 - R = 1 - 0.02 = \boxed{0.98} = 98\%$$

23.8] (a) air because $n_{\text{air}} < n_{\text{water}}$

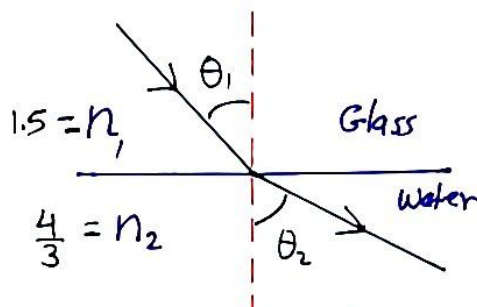
(b) Water, i.e., light incident from water.

23.12] $n_1 \sin \theta_1 = n_2 \sin \theta_2$

$$\sin \theta_2 = \frac{n_1}{n_2} \sin \theta_1$$

$$= \frac{1.5}{4/3} \sin 45^\circ = 0.795$$

$$\Rightarrow \theta_2 = \sin^{-1}(0.795) = \boxed{52.7^\circ}$$

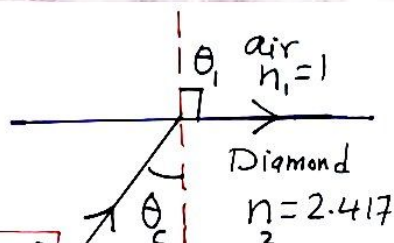


Note: $n_1 > n_2 \Rightarrow \theta_2 > \theta_1$

23.18] $n_1 \sin \theta_1 = n_2 \sin \theta_2$

$$1 \times \sin 90^\circ = 2.417 \sin \theta_c$$

$$\Rightarrow \sin \theta_c = \frac{1}{2.417} \Rightarrow \theta_c = \sin^{-1}\left(\frac{1}{2.417}\right) = \boxed{24.4^\circ}$$

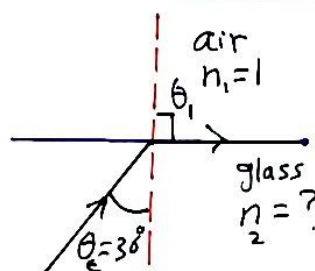


23.20] $n_1 \sin \theta_1 = n_2 \sin \theta_2$

at $\theta_1 = 90^\circ \rightarrow \theta_2 = \theta_c$

$$1 \times \sin 90^\circ = n_2 \sin \theta_c = n_2 \sin 36^\circ$$

$$\Rightarrow n_2 = \frac{1}{\sin 36^\circ} = \boxed{1.7}$$



23.22] $d = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}$

$$d \sin \theta = m \lambda$$

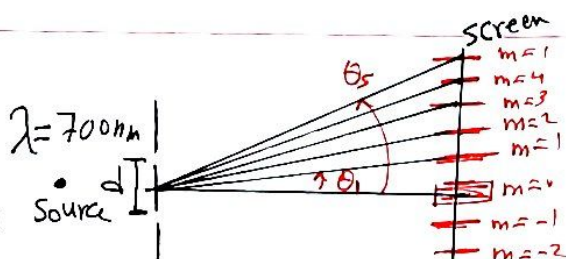
$$\sin \theta = \frac{\lambda}{d} m = \frac{700 \times 10^{-9}}{2 \times 10^{-4}} m = 3.5 \times 10^{-3} m$$

$$\Rightarrow \boxed{\sin \theta = 3.5 \times 10^{-3} m}$$

$$m=1 \Rightarrow \sin \theta_1 = 3.5 \times 10^{-3} \Rightarrow \theta_1 = \sin^{-1}(3.5 \times 10^{-3}) = \boxed{0.2^\circ}$$

$$m=2 \Rightarrow \sin \theta_2 = 2 \times 3.5 \times 10^{-3} \Rightarrow \theta_2 = \boxed{0.4^\circ} \text{ and so on}$$

Note $\theta=0$ for $m=0$



red are bright fringes

2.23]

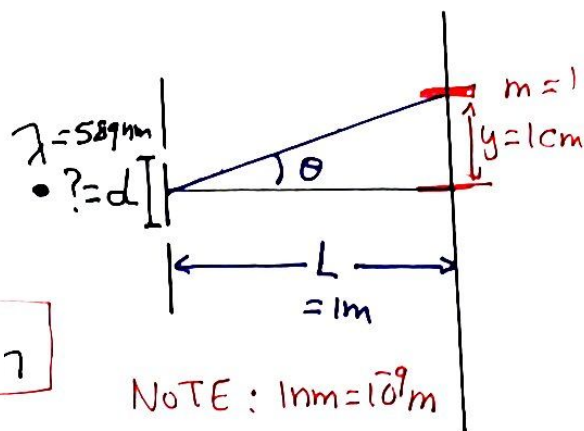
$$1 \text{ cm} = 10^{-2} \text{ m}$$

$$d \sin \theta = m \lambda \Rightarrow \lambda, m=1$$

$$\sin \theta \approx \tan \theta = \frac{y}{L}$$

$$\Rightarrow d \frac{y}{L} = \lambda \Rightarrow d = \frac{\lambda L}{y}$$

$$d = \frac{589 \times 10^{-9} \times 1}{10^{-2}} = \boxed{5.89 \times 10^{-5} \text{ m}}$$



$$2.26] \quad d = \frac{1 \text{ cm}}{5000} = 2 \times 10^{-4} \text{ cm} = 2 \times 10^{-6} \text{ m}$$

$$d \sin \theta = m \lambda \Rightarrow \sin \theta = \frac{m \lambda}{d} = \frac{1 \times 589 \times 10^{-9}}{2 \times 10^{-6}} = 0.2945$$

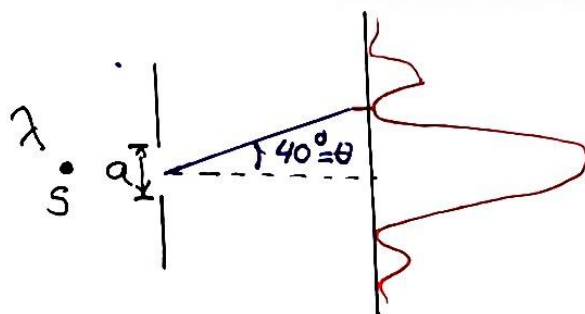
$$\theta = \sin^{-1}(0.2945) = \boxed{17^\circ}$$

$$2.35] \quad \lambda = 589 \text{ nm}$$

$$a \sin \theta = \lambda$$

$$a = \frac{\lambda}{\sin \theta} = \frac{589 \text{ nm}}{\sin 40^\circ}$$

$$\boxed{a = 916 \text{ nm}}$$



$$2.38] \quad R = 10^{-5} \text{ m} \Rightarrow D = 2 \times 10^{-5} \text{ m}$$

$$D \sin \theta = 1.22 \lambda$$

$$\sin \theta = \frac{1.22 \lambda}{D} = \frac{1.22 \times 600 \times 10^{-9}}{2 \times 10^{-5}}$$

$$\theta = \sin^{-1}\left(\frac{1.22 \times 6 \times 10^{-7}}{2 \times 10^{-5}}\right) = \boxed{1.75^\circ}$$

